A Simplified Proof of Weierstrass's Continuous Non-Differentiable Function

Ben Barton-Foskett

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1 Introduction

This paper is meant as a simplified version of Titchmarsh's proof that Weierstrass's function, to be denoted $W(x)$:

$$W(x) = \sum_{n=0}^{\infty} b^n \cos(a^n \pi x)$$

Is continuous for all $x \in \mathbb{R}$. We will begin with key definitions and Theorems for the proof, and the paper will conclude with the simplified proof. Please note this is not my original proof, this is purely meant as an explanation for anyone curious as to how to prove that a function can be continuous everywhere and differentiable nowhere.

2 Prerequisites

This paper assumes prior knowledge of basic results for convergent sequences and series, along with basic ideas of continuity and differentiation from an Analysis I/II class.

3 Uniform Convergence

The proof uses ideas of uniform convergence, so we will first define what it means for a sequence or series of functions to be uniformly convergent.

Definition 3.1. *Let $\sum_{n=0}^{\infty} u_n(x)$ be a series of functions. This series is said to be* uniformly convergent *over the interval (a, b) if:*

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} s.t. \forall n > N, \forall x \in (a, b), |S(x) - S_n(x)| < \varepsilon$$

Where $S_n(x)$ denotes the n-th partial sum of the series. We call a function f uniformly convergent if it uniformly converges over any interval.

Remark 3.2. *If $\sum_{n=0}^{\infty} u_n(x)$ is uniformly convergent, then it is convergent for every value of $x \in (a, b)$.*

It is worth noting that convergence is a property related to an interval or a single point, whereas uniform convergence is a property solely related to intervals. In order to prove uniform convergence, we need to define the *Weierstrass M-Test*, which requires the *Cauchy Convergence Test*:

Lemma 3.3 (Cauchy Convergence Test). *Let $\sum_{n=0}^{\infty} a_i$ be a series. This series is convergent if and only if:*

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} s.t. \forall m > n > N, \left| \sum_{k=n}^{n+m} a_k \right| < \varepsilon$$

Proof. Note that $\sum_{k=0}^{\infty} a_k = \lim_{k\to\infty} S_k$, where S_k is the k-th partial sum of the series. It is known that a sequence is convergent iff it is Cauchy, so (S_k) converges iff it is Cauchy. □

We may now state and prove the Weierstrass M-Test:

Theorem 3.4 (Weierstrass M-Test). *Let (f_n) be a sequence of functions. Also define (M_n) as a sequence of non-negative numbers such that:*

- $|(f_n(x))| \leq M_n$ *for all* $n \geq 1$ *and* $x \in \mathbb{R}$
- $\sum_{n=0}^{\infty} M_n$ *converges*

Then $\sum_{n=0}^{\infty} f_n(x)$ converges absolutely and uniformly.

Proof. Define $S_n(x) = \sum_{k=1}^{n} f_k(x)$. Since $\sum_{n=1}^{\infty} M_n$ converges and $M_n \geq 0 \; \forall n \in \mathbb{N}$, then by 3.3:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m > n > N, \sum_{k=n+1}^{m} M_k < \varepsilon$$

Then, for the chosen N, $\forall x \in \mathbb{R}, \forall m > n > N$:

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^{m} f_k(x) \right| \leq \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} M_k < \varepsilon$$

This shows $S_n(x)$ is Cauchy, and by the completeness axiom, $S_n(x) \to S(x)$. Let $n > N$, so:

$$|S(x) - S_n(x)| = \left| \lim_{m\to\infty} S_m(x) - S_n(x) \right| = \lim_{m\to\infty} |S_m(x) - S_n(x)| \leq \varepsilon$$

Note that N does not depend on x, so (S_n) converges uniformly to the function S. Hence, by definition $\sum_{n=0}^{\infty} f_n(x)$ converges uniformly. Similarly, it can be proven that $\sum_{n=0}^{\infty} |f_n(x)|$ converges uniformly. □

It will next be useful to relate ideas of uniform convergence to continuity, as we will prove that $W(x)$ is continuous through uniform convergence. This leads us to this theorem:

Theorem 3.5 (Uniform Convergence Theorem). *Let (f_n) be a sequence of functions which converge uniformly to f. Also let each f_n be continuous. Then, f is continuous.*

Proof. Let $\varepsilon > 0$. Due to uniform convergence, we can find an $N \in \mathbb{N}$ such that for $n \geq N$:

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3}$$

Also since all f_n are continuous, we can find a $\delta > 0$ such that for $|a - x| < \delta$:

$$|f_N(a) - f_N(x)| < \frac{\varepsilon}{3}$$

Which together gives:

$$\begin{aligned} |f(a) - f(x)| &\leq |f(a) - f_N(a)| + |f_N(a) - f(a)| + |f_N(x) - f(x)| \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &< \varepsilon \end{aligned}$$

Only if $|a - x| < \delta$, giving rise to continuity for f at a. As a was arbitrary we can conclude that uniform convergence implies continuity. □

4 Continuity of Weierstrass's Function

We will here begin the first part of the proof, namely proving that $W(x)$ is everywhere continuous by showing that it is uniformly convergent over any interval. We first recall the definition of the Weierstrass function, which we shall state formally here:

Definition 4.1 (The Weierstrass Function)**.** *We define the Weierstrass Function as:*

$$W(x) = \sum_{n=0}^{\infty} b^n \cos(a^n \pi x)$$

With the following constraints:

- $b \in (0, 1)$
- $a = 2m + 1$, $m \in \mathbb{Z}$

And we shall also state formally that this function is continuous:

Proposition 4.2. *$W(x)$ is uniformly convergent over every interval, thus it is continuous everywhere*

Proof. We first begin by noting that $b \in (0, 1)$, which is equivalent to saying $0 < b < 1$. We also note the following:

$$|b^n \cos(a^n \pi x)| \leq b^n$$

Since $|cos(a^n \pi x)| \leq 1$.
As $0 < b < 1$, we know that $\sum_{n=0}^{\infty} b^n$ converges as a geometric series. Since $\sum_{n=0}^{\infty} b^n$ converges, then by 3.4, $W(x)$ is convergent □

5 Non-Differentiability of Weierstrass's Function

Next, we will prove that $W(x)$ is differentiable nowhere. Inserted here is an extract from Titchmarsh's "The Theory of Functions" as an introduction to this proof:

"If $ab > 1$, the series obtained by term-by-term differentiation is divergent. This in itself does not prove that $f(x)$ is not differentiable, but it suggests possibilities in this direction"

Titchmarsh then goes on to consider the case where $ab > 1 + \frac{3}{2}\pi$, and aims to show that $W(x)$ has no finite differential coefficient for all x. This implies the derivative cannot be explicitly written, thus $W(x)$ is not differentiable.

On first glance, this constraint of $ab > \frac{3}{2}\pi$ seems very random, so we shall gain some intuition behind this before commencing the proof.
Consider the k-th term of the series, namely:

$$b^k \cos(a^k \pi x)$$

Note this function on its own is differentiable, with derivative:

$$\Big(\Big(b^k \cos(a^k \pi x)\Big)\Big)' = -b^k a^k \pi \sin(a^k \pi x)$$

Geometrically, this is a cosine wave with amplitude b^k and frequency a^k. Hence the slope of this wave is roughly $a^k b^k = (ab)^k$, which is roughly proportional to the size of the derivative, as seen through the derivative of the function having a coefficient of magnitude $\pi(ab)^k$. Therefore, the key to $W(x)$ being differentiable depends on how controlled the slopes of each of these cosine waves is in reference to the whole function.
It can then be clear to see that if $ab < 1$ the magnitude of these slopes tend to 0 as $k \to \infty$, so higher frequency waves have less effect on the resulting function, leaving differentiability undisrupted.
However, if $ab > 1$, the magnitude of the slopes grow exponentially, leading us to think that there may be instability in the derivative difference quotient.

While it is possible to prove this for $ab > 1$, we will prove it for $ab > 1 + \frac{3}{2}\pi$, which is sufficient enough for a counter example to disprove the statement *"if a function is continuous then it is differentiable"*, so we now begin the proof.

Proof. We aim to prove that $W(x) = \sum_{n=0}^{\infty} b^n \cos(a^n \pi x)$ is differentiable nowhere for $ab > 1 + \frac{3}{2}\pi$.
First, consider:

$$\begin{aligned}
\frac{W(x+h) - W(x)}{h} &= \sum_{n=0}^{\infty} \frac{b^n \cos(a^n \pi (x+h)) - b^n \cos(a^n \pi x)}{h} \\
&= \sum_{n=0}^{\infty} b^n \frac{\cos(a^n \pi (x+h)) - \cos(a^n \pi x)}{h} \\
&= \sum_{n=0}^{m-1} b^n \frac{\cos(a^n \pi (x+h)) - \cos(a^n \pi x)}{h} + \sum_{n=m}^{\infty} b^n \frac{\cos(a^n \pi (x+h)) - \cos(a^n \pi x)}{h} \\
&= S_m + R_m
\end{aligned}$$

What we have done here is further explored the ideas above through splitting the potential derivative of $W(x)$ into the low and high frequency waves, at an arbitrary index $m \in \mathbb{N}$. Namely, S_m is the low frequency waves and R_m is the high frequency waves. We now aim to show that R_m will become increasingly large.
Next, we shall use the mean value theorem to bound the numerator of the difference quotient. The mean value theorem states that since $\cos(a^n \pi (x+h)) - cos(a^n \pi x)$ is differentiable, $\exists \theta \in (0,1)$ such that:

$$\begin{aligned}
\cos(a^n \pi (x+h)) - cos(a^n \pi x) &= h \cdot (\cos(a^n \pi (x + \theta h)))' \\
&= -h a^n \pi \sin(a^n \pi (x + \theta h))
\end{aligned}$$

Which allows us to bound the numerator as follows:

$$|\cos(a^n \pi (x+h)) - cos(a^n \pi x)| = |a^n \pi h \sin(a^n \pi (x + \theta h)| \leq a^n \pi |h|$$

Letting s_m be the m-th term of S_m and dividing everything by $|h|$ yields:

$$\begin{aligned}\frac{|\cos(a^n\pi(x+h)) - cos(a^n\pi x)|}{|h|} &\leq \pi a^n \\ \frac{|s_m|}{b^n} &\leq \pi a^n \\ |s_m| &\leq \pi a^n b^n\end{aligned}$$

Therefore, by taking the sum of both sides we obtain:

$$|S_m| \leq \sum_{n=0}^{m-1} \pi a^n b^n$$

And note that the term on the right of the inequality is a geometric sum, with common ratio ab, and a geometric sum can be written as $\sum_{n=0}^{k} r^k = \frac{r^k-1}{r-1}$, so the inequality becomes:

$$|S_m| \leq \pi \frac{a^m b^m - 1}{ab - 1} < \pi \frac{a^m b^m}{ab - 1} \qquad (*)$$

We next look to find a lower bound for R_m using a particular value for h. We shall first begin by considering $a^m x$ by splitting it into its integer part and real remainder:

$$a^m x = \alpha_m + \xi_m$$

For $\alpha_m \in \mathbb{Z}$ and $\xi_m \in \left[-\frac{1}{2}, \frac{1}{2}\right)$. Now, somewhat arbitrarily we will choose a value of h that will force the cosine function to produce ± 1, leading to a divergent series. This value of h will need to have the property that $h \to 0$ as $a^m \to \infty$. We will choose h as follows:

$$h = \frac{1 - \xi_m}{a^m}$$

Note that this h has the property we required. We also note the upper and lower bounds to this h based on $-\frac{1}{2} \leq \xi_m < \frac{1}{2}$:

$$0 < h \leq \frac{3}{2a^m} \qquad (**)$$

Now we consider the term inside the cosine of $W(x+h)$ to see how it simplifies based on this choice of h:

$$\begin{aligned}a^n\pi(x+h) &= a^{n-m} \cdot a^m \pi(x+h) \\ &= a^{n-m} \cdot a^m \pi(x + \frac{1-\xi_m}{a^m}) \\ &= a^{n-m} \cdot a^m \pi(\frac{a^m x + 1 - \xi_m}{a^m}) \\ &= a^{n-m}\pi([a^m x - \xi_m] + 1) \\ &= a^{n-m}\pi(\alpha_m + 1)\end{aligned}$$

We now consider this term inside the cosine, and recall that a is odd:

$$\begin{aligned}\cos(a^n\pi(x+h)) &= \cos(a^{n-m}\pi(\alpha_m + 1)) \\ &= (-1)^{a^{n-m}(\alpha_m+1)} \\ &\text{As } a \text{ is odd} \\ &= (-1)^{\alpha_m+1}\end{aligned}$$

Next, we consider $W(x)$, using similar reasoning as before:

$$\begin{aligned}\cos(a^n\pi x) &= \cos(a^{n-m}\pi(\alpha_m + \xi_m)) \\ &= \cos(a^{n-m}\pi\alpha_m)\cos(a^{n-m}\pi\xi_m) \\ &= (-1)^{\alpha_m}\cos(a^{n-m}\pi\xi_m)\end{aligned}$$

So we can conclude for R_m:

$$R_m = \frac{(-1)^{\alpha_m+1}}{h}\sum_{n=m}^{\infty} b^n(1+\cos(a^{n-m}\pi\xi_m))$$

Note that for the series, all the terms are positive, so by considering only the first term ($n = m$), and using $(**)$:

$$|R_m| > \frac{b^m}{|h|} > \frac{2}{3}a^m b^m$$

Thus, we have found a lower bound for R_m.
Now we put this together with $(*)$ to obtain:

$$\left|\frac{W(x+h)-W(x)}{h}\right| \geq |R_m| - |S_m| > \left(\frac{2}{3} - \frac{\pi}{ab-1}\right)a^m b^m$$

Then we can note if $ab > 1 + \frac{3}{2}\pi$, the quantity in the brackets is positive, and as $m \to \infty$, $h \to 0$ and the whole right hand side of the inequality tends to infinity. Therefore the difference quotient takes arbitrarily large values, i.e. $W(x)$ has no finite differential coefficient, so $W'(x)$ either cannot exist, or it is not finite. In either case, it has been shown the $W(x)$ is not differentiable anywhere. This concludes the proof □

6 References

- Edward Charles Titchmarsh (1939). The theory of functions. Second Edition ed. Oxford Oxford Univ. Press, pp.1–8 and 351–353.
- Wachsmuth, B.G. (2018). MathCS.org - Real Analysis: Theorem 8.2.3: Uniform Convergence preserves Continuity. [online] Mathcs.org. Available at: https://mathcs.org/analysis/reals/funseq/proofs/uconvcont.html [Accessed 15 Feb. 2026].